



The real effects of AI: Evidence from corporate investment efficiency[☆]

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ABSTRACT

This study examines how artificial intelligence (AI) adoption is associated with corporate investment efficiency among U.S. firms. We measure AI adoption using *AI mentions*, defined as the proportion of AI-related terminology in senior management remarks during earnings calls, and validate this measure by showing that it is positively associated with 10-K discussions of AI applications in operational decision-making, product development, and business expansion. We construct an instrumental variable based on exposure to university-authored AI research—measured as a normalized score of textual similarity between non-corporate academic publications and industry descriptions—to address concerns that AI adoption may be endogenous to firms' investment decisions and investment efficiency. Our baseline results indicate that higher AI adoption is associated with greater investment efficiency, and this association remains in two-stage least squares regressions using the AI technology exposure instrument. We also find that firms with higher AI mentions tend to have higher Tobin's *q* over longer horizons, consistent with a gradual reflection of AI-related benefits in firm value. Additional analyses suggest several mechanisms: AI adoption is associated with more accurate management sales forecasts, higher financial reporting quality, and greater process and product innovation intensity. Cross-sectional evidence further indicates that regulatory frictions, such as exposure to data-privacy regulation, attenuate the association between AI adoption and investment efficiency. Overall, the findings suggest that AI adoption is an emerging factor in corporate investment and capital allocation.

1. Introduction

Recent advances in artificial intelligence (AI) are changing how firms process information for real investment decisions. Tools based on machine learning and natural language processing can analyze large, complex datasets and, in some settings, generate risk assessments and predictions that systematically outperform human judgment given the same information (Kleinberg et al., 2018;

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Jansen et al., 2025). Empirical evidence shows that AI-based tools can support data-driven decision-making and generate systematic corporate policy recommendations (Brynjolfsson and McElheran, 2016; Campello et al., 2025). At the same time, studies document that firms investing in AI or related general-purpose digital technologies affect core business and investment policies rather than only support functions (Brynjolfsson and Hitt, 2000; Bloom et al., 2021; Fedyk et al., 2022; Babina et al., 2024a). However, it is not clear ex ante whether these technologies lead to measurably more efficient investment within firms.

A large literature emphasizes that information quality and managerial forecasting ability are important determinants of investment efficiency. Classic theories link information asymmetry and agency conflicts to deviations from optimal investment, while richer information environments help align financing and investment with underlying fundamentals (Jensen and William, 1976; Myers and Nicholas, 1984; Biddle et al., 2009). Empirical work further shows that managers with stronger forecasting and planning capabilities undertake investments and acquisitions closer to value-maximizing benchmarks, exhibiting more efficient capital expenditures and higher-quality deals (Goodman et al., 2014; Tanaka et al., 2020). Complementing this evidence, Gondhi (2023) shows, within a rational inattention framework, that managers' limited information-processing capacity and attention allocation lead to resource misallocation and lower aggregate output, even when decisions are otherwise rational. Against this backdrop, AI can be viewed as a technology that can improve investment efficiency by enhancing managerial forecasting, supporting financial reporting quality, and expanding firms' capacity to discover and evaluate innovative projects.

Realizing these potential benefits, however, is neither immediate nor costless. Effective use of AI typically requires complementary investments in data infrastructure, human capital, and organizational redesign, and these intangible outlays can be sizable and front-loaded (Brynjolfsson and Hitt, 2000; Brynjolfsson et al., 2021). Moreover, the diffusion of disruptive technologies such as AI is gradual and uneven across regions, firms, and occupations, and complementary capabilities and adoption costs shape when and where AI gains materialize (Bloom et al., 2021). AI systems can also introduce new sources of risk—including opacity, overfitting, and sensitivity to data quality—that may lead to suboptimal portfolio choices or regulatory concerns in financial applications (D'Acunto et al., 2019; Campello et al., 2025). Experimental evidence further shows that AI agents can generate hallucinations, agency-like problems, or costly “agreement spirals” when tools and oversight are imperfect, highlighting additional operational risks from AI deployment (Canayaz, 2025). In addition, AI-related standards and governance can amplify or dampen AI investment and capital budgeting, and AI adoption has been associated with higher systematic risk and changes in firms' cost of capital; long-run effects appear to depend on how well AI is integrated with existing human and organizational capabilities (Babina et al., 2024b; Cao et al., 2024; Canayaz and Wang, 2025). In this setting, AI-related risks, adjustment costs, and implementation frictions represent costs that must be weighed against the potential gains from improved information processing and prediction.

Taken together, AI adoption may improve investment efficiency by enhancing information and decision-making, but it may also introduce new risks and require sizable complementary investments. Whether AI adoption is associated with higher investment efficiency is therefore an empirical question. Using a sample of 16,145 firm-year observations for 2401 U.S. firms from the StreetEvents–Compustat–CRSP universe over 2010–2021, we examine how firms' AI adoption relates to investment efficiency.

To measure AI adoption, we construct a firm-year measure based on the frequency of AI-related terminology in earnings call transcripts, serving as a proxy for senior executives' stated strategic focus on AI. Specifically, we define *AI mentions* as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Because there is no widely accepted, comprehensive AI keyword dictionary, we develop our own using ChatGPT-assisted term generation and manual textual analysis, following keyword-discovery and dictionary-construction methodologies in recent work (Sautner et al., 2023; Eisfeldt et al., 2026). This measure allows us to capture two aspects of firms' use of AI in investment decisions: the extensive and intensive margins of AI applications. The extensive margin involves expanding activities, discovering new opportunities, or increasing the pool of options available to a firm, whereas the intensive margin reflects the extent to which AI technologies may help firms select better projects from their existing opportunity set.¹ We validate the measure by examining its association with AI-related discussions in 10-K filings. *AI mentions* is positively correlated with discussions of AI applications aimed at enhancing decision-making in existing operations and expanding investment opportunities. It is also significantly associated with whether firms describe intensive (process-focused) and extensive (opportunity-expanding) AI benefits in their 10-K disclosures, suggesting that it captures meaningful variation in AI adoption related to investment decision-making.

Our main analysis indicates that AI adoption is positively associated with greater investment efficiency, measured as firms' deviations from expected investment given their growth opportunities, after controlling for observable firm characteristics. A one-standard-deviation increase in *AI mentions* is associated with a 2.17% (3.65%) increase in investment efficiency at the mean (median). Given the potential for endogeneity in AI adoption, we estimate a two-stage least squares (2SLS) specification that uses exposure to university-authored AI research—measured by textual similarity between academic AI publications and industry descriptions—as an instrumental variable for *AI mentions*, in the spirit of Chen and Srinivasan (2024). This instrument is strongly correlated with firms' propensity to discuss AI, satisfying the relevance condition. Because we filter out all corporate coauthorships, the instrument captures knowledge spillovers arising exclusively from academic research rather than from firm-specific investment decisions, thereby mitigating—though not eliminating—concerns about violations of the exclusion restriction. Our 2SLS estimates are consistent with the

¹ Examples include AstraZeneca plc, which uses generative AI to help researchers interrogate information, identify untapped areas for drug development, and broaden its research portfolio—an illustration of the extensive margin of AI applications (“Generative AI promises an economic revolution. Managing the disruption will be crucial,” The Wall Street Journal, August 28, 2023)—and BMW Group, which employs AI technologies to optimize its existing manufacturing and inspection processes—an illustration of the intensive margin (“How BMW uses A.I. to make vehicle assembly more efficient,” CNBC, July 21, 2023).

baseline results, although we interpret them with caution given the untestability of the exclusion restriction.

Because AI adoption appears to be associated with improved capital allocation, we examine whether its benefits are reflected in firm performance. We find that firms with higher *AI mentions* tend to have higher Tobin's q over one- to three-year horizons, consistent with gradual increases in firm value as firms adjust and invest in complementary assets. Analyses of stock returns indicate generally positive, but smaller and less consistent, associations with *AI mentions*, suggesting that equity markets may incorporate expected benefits relatively quickly while valuation effects become more pronounced over time.

Next, we examine potential mechanisms through which AI adoption relates to investment efficiency. First, we study whether AI adoption is associated with more accurate management sales forecasts. AI may enhance managerial prediction by enabling analysis of large-scale datasets beyond traditional human judgment, helping firms better align investment with growth opportunities and reduce costly misallocations. In line with this view, we find that higher *AI mentions* is associated with significantly lower management sales forecast errors over multi-year horizons. Additional tests suggest that the mechanism operates primarily through managers' internal use of AI rather than through improvements in analyst forecast accuracy or reductions in firm risk.

Second, we examine whether AI adoption improves the quality of external financial reporting by constraining earnings management. To the extent that AI-enabled systems increase the internal visibility of accounting irregularities, they may raise the expected cost of earnings management and thereby reduce information asymmetry between managers and outside investors. Consistent with this view, we find that higher *AI mentions* is associated with lower real and accrual-based earnings management. These results suggest that AI adoption may contribute to higher investment efficiency in part through improved financial reporting quality, even though its effects on conventional operating efficiency measures appear limited in our setting.

Third, we analyze patent activity as an additional mechanism. AI may lower the cost of evaluating and screening new projects, enabling firms to allocate capital toward higher-quality opportunities. We find that higher *AI mentions* is associated with greater intensity of patenting activity, measured as process and product patents scaled by sales, suggesting that AI helps managers identify and develop efficiency-enhancing process improvements and new products, potentially contributing to more effective capital allocation over time.

We conduct several robustness tests to validate our findings. First, our subsample analyses of early and later AI adoption periods provide limited evidence that the positive association between AI adoption and investment efficiency is concentrated among early adopters. Second, we use the General Data Protection Regulation (GDPR) as a regulatory shock and find that the positive association between *AI mentions* and investment efficiency weakens for firms more exposed to the European Union (EU) after its adoption. Third, we address concerns about time-varying, industry-level shocks by using industry-year fixed effects rather than year fixed effects and find that our main inferences remain similar. Fourth, we decompose AI-related terminology into five subfields and find that discussions of computing and hardware infrastructure, and to a lesser extent data and analytics tools in the initial period, are more strongly associated with investment efficiency than other AI subfields. Fifth, we decompose investment into capex and non-capex components and find that *AI mentions* is particularly associated with improved efficiency in R&D-related investment, consistent with AI facilitating the allocation of innovative capital over time. Sixth, we distinguish between underinvestment and overinvestment and find that AI adoption tends to improve investment efficiency for firms prone to underinvestment. Lastly, we use the share of AI specialists in the high-skilled workforce as an alternative proxy for AI adoption and obtain results that are broadly consistent with our baseline.

Our study contributes to the literature in several ways. First, it extends research on the diffusion of new technologies and their effects on economic growth and firm valuation (Bartel et al., 2007; Comin and Mestieri, 2014; Brynjolfsson and McElheran, 2016). Recent work provides further evidence on how AI investments and exposure affect firm value, systematic risk exposure, workforce outcomes, and AI-based prediction of corporate performance (Babina et al., 2024a; 2024b; Campello et al., 2025; Eisfeldt et al., 2026). Our research examines the role of AI in corporate decision-making by analyzing how AI technologies may help firms identify new projects and allocate capital more efficiently within their existing opportunity sets.

Second, our study contributes to the literature by examining how AI may help address constraints that distort the investment process, such as information asymmetry and limited managerial capacity (Harris and Raviv, 1996; Jagannathan et al., 2016). Our analysis suggests that AI adoption is associated with improved investment decision-making, as managers may use AI-based analytical tools to assist in selecting and valuing investment projects. By quantifying AI adoption and its association with investment efficiency, we provide evidence on an important performance metric for corporate stakeholders.

Finally, our study contributes to research on how new technologies transform human decision-making in specific contexts, such as judicial bail decisions (Kleinberg et al., 2018), robo-advising for retail investors in India (D'Acunto et al., 2019), auditing processes (Fedyk et al., 2022), and algorithmic underwriting (Jansen et al., 2025). In contrast to these studies, which focus on discrete decision problems in law, personal finance, auditing, or consumer credit, we investigate how AI adoption influences corporate investment decision-making and its broader associations with capital allocation and firm performance. We also highlight AI's potential to complement, rather than replace, human judgment in high-skill occupations, as evidenced by research on AI-based tools that enhance financial decision-making and productivity (Grennan and Michaely, 2021; Noy and Zhang, 2023). Our findings suggest that AI adoption is associated with more accurate management forecasts and more efficient investment decisions.

The remainder of the paper is organized as follows. Section 2 reviews the related literature and develops the main hypothesis. Section 3 describes the data, variable construction, and key measures of *AI mentions* and investment efficiency. Section 4 presents the main analysis of AI adoption, investment efficiency, and firm value. Section 5 examines potential mechanisms underlying these associations, while Section 6 provides additional tests. Section 7 concludes.

2. Literature review and hypothesis development

2.1. AI adoption by firms

AI adoption has become more common across industries, prompting growing scholarly inquiry into its effects on firm performance, labor markets, decision-making, and sector-specific outcomes. Studies document that AI and related digital tools can affect operational efficiency and competitive positioning, including in financial services (D'Acunto et al., 2019; Abis and Veldkamp, 2024). For example, D'Acunto et al. (2019) examine how a robo-advisory optimizer changes portfolio diversification, risk-taking, and behavioral biases among retail investors at a large Indian broker, while Abis and Veldkamp (2024) analyze how AI-driven data analysis reshapes information production and business models in investment management and other data-intensive services. Babina et al. (2024a) show that firms with greater AI investment experience faster growth in sales, employment, and market value, primarily through increased product innovation rather than pure process-cost reductions. A growing body of research finds that decision quality can improve when algorithmic or AI-based tools complement human judgment, especially in complex information environments (Kleinberg et al., 2018; Cao et al., 2024). In the venture capital context, Lyonnet and Stern (2025) show that combining human expertise with algorithmic analytics is associated with more accurate market assessments and greater innovation among portfolio firms. AI adoption is also associated with changes in labor market dynamics, shifting the capital-labor mix and reallocating demand toward certain skilled occupations while displacing tasks in others (Abis and Veldkamp, 2024; Eisfeldt et al., 2026). The need for new skill sets and organizational adaptation thus increases as firms broaden their use of AI. Taken together, these studies suggest that AI adoption is associated with improved decision quality and performance, motivating our focus on investment efficiency.

2.2. AI adoption and investment efficiency

Accurate forecasting plays an important role in shaping firms' real investment decisions. Improved prediction can reduce financing frictions and agency-driven deviations from value-maximizing investment by alleviating information asymmetry (Jensen and William, 1976; Myers and Nicholas, 1984). While higher forecast accuracy is associated with more efficient investment outcomes (Goodman et al., 2014; Tanaka et al., 2020), limited managerial information-processing capacity and attention allocation can contribute to resource misallocation even when choices are otherwise rational (Gondhi, 2023). AI adoption may directly address these informational and cognitive frictions: recent advances in machine learning and natural language processing enable firms to analyze large-scale datasets and improve forecasting accuracy in ways that go beyond traditional human judgment (Kleinberg et al., 2018; Campello et al., 2025), and algorithmic models have been shown to outperform human decision-makers in complex prediction tasks (Kleinberg et al., 2018; Jansen et al., 2025), suggesting that AI-supported managerial forecasting may meaningfully reduce the information gaps that lead to over- or underinvestment.

AI adoption may also improve investment efficiency through the quality of external financial reporting. High-quality financial reporting reduces information asymmetry between managers and outside investors, enabling efficient capital allocation (Richardson, 2006; Biddle et al., 2009). However, reporting quality can be undermined by earnings management, whereby managers exercise discretion over accruals or real activities to misrepresent firm performance (Cohen et al., 2008; Cohen and Zarowin, 2010). AI-enabled systems may constrain such manipulation by automating transaction monitoring and anomaly detection across large volumes of financial data, raising the visibility of accounting irregularities and the expected cost of opportunistic manipulation. Consistent with this view, Fedyk et al. (2022) find that AI adoption in audit firms reduces the incidence of restatements related to accruals and revenue recognition. To the extent that AI adoption reduces earnings management and improves reporting quality, it may further support investment efficiency.

AI adoption may further foster innovation in both product development and production processes, which can directly shape firms' investment opportunities and capital allocation decisions (Babina et al., 2024a). Cockburn et al. (2018) argue that AI serves as a general-purpose method of invention, enabling firms to leverage large datasets and learning algorithms to open up new approaches to R&D and accelerate the discovery of commercially viable projects. In particular, AI may help managers assess new projects more effectively and screen potential investments, reducing the cost of experimentation and enabling firms to commit capital to higher-quality opportunities. Process innovations, which involve new methods of production, can facilitate adjustments in firms' input mix and protect firm value (Bena et al., 2022), while product innovations expand firms' investment opportunity set. To the extent that AI adoption promotes both types of patenting activity, it may further contribute to more effective capital allocation.

At the same time, these potential benefits are accompanied by nontrivial costs. Regulatory limits on data access can weaken predictive performance, and adopting AI systems often requires substantial investments in staff training, data infrastructure, and process changes (Brynjolfsson and McElheran, 2016; Brynjolfsson et al., 2021; Aridor et al., 2023). AI models may also introduce risks, including overfitting, opacity, and sensitivity to poor-quality or incomplete data, which can in turn create scope for erroneous decisions or regulatory concerns (D'Acunto et al., 2019; Campello et al., 2025; Canayaz, 2025; Canayaz and Wang, 2025). Investments in AI are further associated with higher exposure to systematic risk, which can raise firms' cost of capital (Babina et al., 2024b), and realizing long-term gains may depend on complementary investments (Brynjolfsson and Hitt, 2000; Fedyk et al., 2022; Cao et al., 2024). As a result, the gains from AI adoption may be partly offset by regulatory, organizational, and model-related frictions.

In sum, AI adoption may improve investment efficiency through improved forecast accuracy, higher financial reporting quality, and greater innovation activity. However, implementation costs, regulatory constraints, and modeling risks may limit or counteract these benefits. Thus, the overall effect of AI adoption on investment efficiency is an empirical question.

Hypothesis (stated as the null): *Firms with higher levels of AI adoption do not differ in investment efficiency compared to other firms.*

3. Data and variable definitions

3.1. Sample

Our sample spans the period from 2010 to 2021, during which demand for AI-skilled jobs increased significantly (Babina et al., 2024a). Table 1 summarizes the sample construction. We begin with 72,124 firm-year observations and 630,468 quarterly earnings call transcripts from the Refinitiv StreetEvents database, including non-U.S. firms. We then exclude 38,594 firm-year observations not found in either Compustat or CRSP, 7027 firm-year observations in financial industries (Standard Industrial Classification (SIC) codes 6000–6999), and 992 firm-year observations in utility industries (SIC codes 4900–4999), because investment for regulated firms differs systematically from that of other firms (Biddle et al., 2009). Following Acemoglu et al. (2022) and Babina et al. (2024a), we also exclude 4587 firm-year observations in AI-innovating industries (North American Industry Classification System (NAICS) codes 51 and 54) to focus on firms that primarily use AI rather than develop AI technologies. Finally, we exclude 4779 firm-year observations with missing values for key variables. After applying these criteria, our final sample consists of 2401 unique U.S. firms, yielding 16,145 firm-year observations in the merged StreetEvents–Compustat–CRSP dataset.²

3.2. Measure of AI adoption: AI mentions in earnings calls

Our main proxy for firm-level AI adoption is constructed from earnings conference call transcripts and is based on executives' statements. We analyze both management presentations and question-and-answer interactions with analysts, investors, and the media.³ The text is pre-processed using standard natural language processing techniques, including tokenization and lowercasing, to handle variation in word forms across calls. We identify sentences from quarterly calls that contain AI-related keywords, initially defined using the dictionary developed by Babina et al. (2024a). We then prompt ChatGPT, a large language model instructed to act as an AI expert, to answer three questions based on excerpts from these transcripts. The first question classifies whether and how a firm discusses AI involvement—distinguishing among active application, investment, strategic focus, and acquisitions. In our sample of 2401 firms, 305 (12.70%) mention AI applications during earnings calls. Follow-up questions elicit brief descriptions of the types, purposes, and strategic areas of AI use. Fig. 1 presents word clouds for three benchmark years—2012, 2016, and 2020—showing changes in AI-related terms over time, with a broader range of keywords appearing in more recent years. Appendix B describes this procedure in detail.

Using these outputs, we compile and filter AI-related keywords, drawing on multiple sources including internet searches and technical glossaries (Appendix C). We then define *AI mentions* as the ratio of AI-related keyword counts used by a firm's senior management team to the total number of words in the transcripts. This measure is intended to capture the intensity with which top managers discuss AI applications, and we interpret it as potentially reflecting management's strategic emphasis on and commitment to AI. To assess classification accuracy, we manually review a subset of transcripts. Finally, we compute each firm's annual average *AI mentions* from the quarterly observations.

To relate our AI vocabulary to standard terminology, we benchmark the Appendix C dictionary against glossaries from AI textbooks (e.g., Russell and Norvig, 2024) and technical standards. This reference vocabulary allows us to assess how closely managerial language aligns with recognized AI terminology. The analysis indicates that executives' language is concentrated in applied domains—such as machine learning, deep learning, computer vision, natural language processing, and predictive analytics—while many specialized technical topics in the reference glossaries (for example, specific learning algorithms or automata theory) do not appear in earnings calls.⁴ This pattern is consistent with evidence that firms and standard-setting bodies emphasize applied AI concepts rather than the full range of technical vocabulary (Canayaz and Wang, 2025).

3.3. Validation of AI mentions

We next examine the properties of *AI mentions* to assess whether it plausibly captures the extent of AI adoption in firms' investment decision-making processes.

3.3.1. AI applications in 10-K business descriptions

We first validate the measure by manually reviewing discussions in earnings conference call transcripts (Appendix D). Firms describe "intensive" AI applications, such as enhanced risk assessment and decision-making for existing operations (e.g., optimized pricing strategies and improved operational efficiency), and "extensive" applications, such as new product development and market expansion. If *AI mentions* reflects a firm's AI adoption in investment decision-making, we would expect more detailed discussions of AI-

² Estimation samples differ across tables because we exclude singleton observations and drop firm-year observations with missing data for the variables used in each specification.

³ The most frequently mentioned executive job titles in earnings calls include CEO, CFO, CIO, COO, executive chair, divisional executives, presidents, vice presidents, and group or regional senior executives.

⁴ In our mapping, about 45% of the 160 unique AI keywords have direct matches in the reference glossary, 35% are conceptually covered by broader entries, and 20% are domain-specific or reflect business infrastructure, platforms, or vendor branding; see Online Appendix Table A.1 for the full mapping.

Table 1
Sample construction and screening process.

Number of firm-year observations in the StreetEvents database from 2010 to 2021 (all firms, including non-U.S. firms)	72,124
Number of quarterly earnings call transcripts in the Refinitiv StreetEvents database from 2010 to 2021	630,468
Less:	
Number of firm-year observations not found in the Compustat and CRSP databases	(38,594)
Number of firm-year observations in financial industries (SIC 6000–6999)	(7027)
Number of firm-year observations in utility industries (SIC 4900–4999)	(992)
Number of firm-year observations in AI-innovating industries (NAICS 51 and 54)	(4587)
Number of firm-year observations with missing values for key variables	(4779)
Final sample	
Number of unique firms in the final StreetEvents–Compustat–CRSP sample	2401
Number of firm-year observations in the final StreetEvents–Compustat–CRSP sample (US firms only)	16,145

This table summarizes the construction of the StreetEvents–Compustat–CRSP sample of 2401 US firms and 16,145 firm-year observations for the period 2010–2021.

related activities and operational strategies in Item 1 of the 10-K filing. To test this, [Appendix E](#) reports correlations between *AI mentions* and variables capturing AI application discussion in 10-K filings. These include the number of sentences describing AI technologies assisting in the selection and management of existing projects (*No. of intensive margin*) and the number describing AI technologies that expand business scope (*No. of extensive margin*). The correlations between *AI mentions* and *No. of intensive margin* and *No. of extensive margin* are 0.234 and 0.241, respectively. The correlations remain positive when these sentence counts are scaled by the total number of AI dictionary sentences ([Appendix C](#)) in the 10-K business description. These patterns suggest that firms with higher *AI mentions* tend to discuss AI applications more extensively in their business activity disclosures.⁵

3.3.2. Intensive and extensive AI application indicators in 10-K filings

We further estimate ordinary least squares (OLS) regressions to provide additional validation for *AI mentions*. Specifically, we analyze Item 1 of the 10-K, which describes a company's business and key strategic activities and is widely used in the literature to study firms' operations and competitive positioning ([Hoberg and Phillips, 2024](#)). Our approach is based on the premise that firms that meaningfully adopt AI technologies are more likely to discuss these technologies and their strategic implications in Item 1.

Using a methodology consistent with that applied to conference calls ([Appendix B](#)), we identify passages that contain AI-related terminology and are explicitly linked to operational or strategic applications. We construct two indicator variables to distinguish between the extensive and intensive margins of AI-related efficiency gains in investment decision-making. *Intensive AI benefits* equals one if the 10-K mentions AI-related keywords or phrases describing how AI technologies assist firms in improving decision-making for existing operations, and zero otherwise. Examples include improving pricing strategies, enhancing inventory management, and implementing continuous product evaluation. *Extensive AI benefits* equals one if the 10-K mentions AI-related keywords or phrases describing how AI technologies help firms expand investment opportunities, and zero otherwise. Examples include launching new products, expanding market reach, and identifying strategic partnerships.

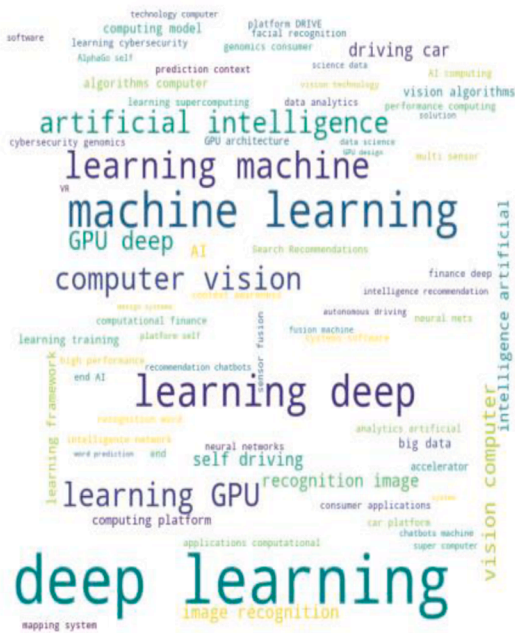
[Appendix F](#) presents the regression results. The dependent variables are *Intensive AI benefits* in column (1) and *Extensive AI benefits* in column (2). In addition to firm-level characteristics, we control for $\text{Log}(1 + 10\text{-K size})$ to account for the effect of report length on the likelihood of discussing AI applications. The coefficients on *AI mentions* are positive and significant at the 1% level, indicating that firms with higher *AI mentions* are more likely to discuss both intensive and extensive AI benefits. For example, a one-standard-deviation increase in *AI mentions* of 0.137 is associated with a 1.51 percentage point (1.30 percentage point) higher probability of discussing AI applications with intensive (extensive) benefits. These changes correspond to 18.57% (16.39%) of unconditional probabilities, suggesting economically meaningful associations.

Taken together, the validation results indicate that *AI mentions* is associated with AI applications across various business operations, including those connected to investment decisions. This supports our interpretation of *AI mentions* as a proxy for AI adoption, capturing both the expansion of investment opportunities and changes in project selection.

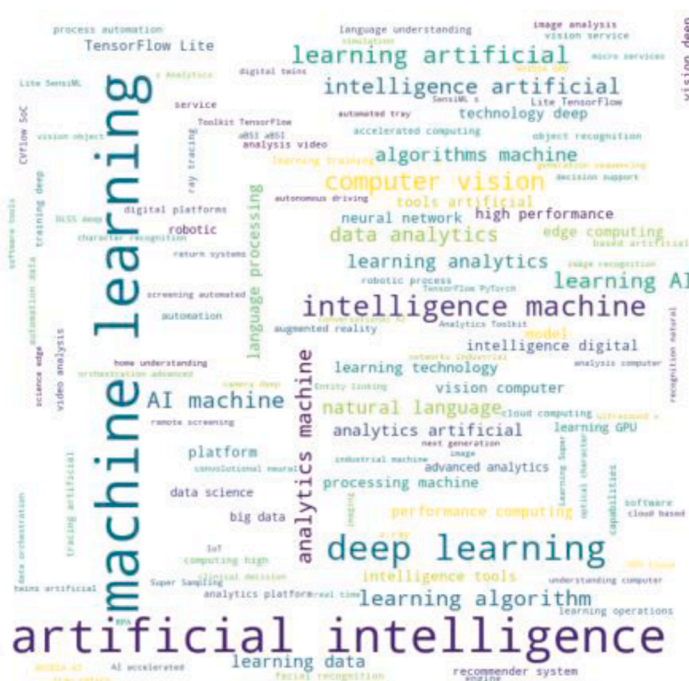
3.4. Descriptive statistics

Panel A of [Table 2](#) reports summary statistics for the main variable as well as for firm and governance characteristics in the StreetEvents–Compustat–CRSP sample. All continuous variables are winsorized at the 1st and 99th percentiles. The key independent variable, *AI mentions*, is scaled by 10^3 for ease of interpretation. The mean (median) *AI mentions* is 0.03 (0.00), reflecting low AI diffusion over the 2010–2021 period. Online Appendix Figure A.1 plots average *AI mentions* over this period, from near zero in 2010 to about 0.04 by 2021. This pattern of low averages with cross-sectional dispersion is broadly consistent with prior research on the

⁵ We employ predefined lexicons developed by [Loughran and McDonald \(2011\)](#) to extract insights on disclosures related to AI applications. Managers leveraging AI technologies may articulate their strategies more specifically, using terms that convey technological enhancements or market opportunities. Firms that extensively integrate AI may discuss regulatory compliance or data-privacy issues more explicitly, reflecting their awareness of legal risks. As shown in the last two rows of [Appendix E](#), we find that firms with higher *AI mentions* use more phrases associated with legal risks ("litigious words") and specific terminology ("precise words") when discussing AI adoption.



(b) AI technology-related keywords in 2016



(c) AI technology-related keywords in 2020

diffusion of transformative technologies (Brynjolfsson and Hitt, 2000; Jorgenson et al., 2008; Comin and Hobijn, 2010; Babina et al., 2024a; Cao et al., 2024). Panel B reports statistics on AI mentions by one-digit SIC code, indicating cross-sectional variation in AI discussions.

Table 2
Descriptive statistics.

Panel A. Summary statistics for <i>AI mentions</i> and firm characteristics							
	Mean	SD	P10	P25	P50	P75	P90
AI mentions ($\times 1000$)	0.025	0.137	0	0	0	0	0
Market capitalization (\$ millions)	5033	10,803	82	334	1135	4456	13,058
Firm age (years)	24.711	17.559	5.507	10.008	20.433	33.025	54.036
Book-to-market	0.469	0.556	0.075	0.199	0.382	0.651	1.009
Leverage	0.262	0.246	0	0.063	0.228	0.382	0.552
Dividend (indicator)	0.449	0.497	0	0	0	1	1
Sales volatility	0.169	0.173	0.037	0.065	0.116	0.209	0.356
Investment volatility	0.081	0.217	0.011	0.022	0.047	0.093	0.16
Asset tangibility	0.284	0.265	0.044	0.093	0.191	0.399	0.69
Z-score	0.968	1.87	−0.556	0.556	1.187	1.806	2.53
Negative income (indicator)	0.316	0.465	0	0	0	1	1
Cash flow volatility	0.074	0.174	0.011	0.018	0.033	0.066	0.147
Cash flow to sales	−0.573	4.969	−0.185	0.037	0.095	0.165	0.271
Cash to PPE	5.531	22.842	0.044	0.161	0.607	2.359	9.053
Operating cycle (days)	32.842	24.023	5.57	16.118	30.645	42.839	57.361
Institutional ownership	0.58	0.382	0	0.148	0.731	0.897	0.979
CEO chair duality (indicator)	0.365	0.481	0	0	0	1	1
Board size	8.712	2.78	6	7	8	10	12
Board independence	0.795	0.119	0.625	0.75	0.833	0.889	0.9
Panel B. <i>AI mentions</i> by one-digit SIC code							
One-digit SIC code	Mean	SD	No of observations				
Agriculture, Forestry, and Fishing	0	0	24				
Mining and Construction	0.007	0.047	1375				
Manufacturing (Food, Textile, and Chemical Products)	0.009	0.061	3894				
Manufacturing (Machinery and Miscellaneous)	0.021	0.095	6346				
Transportation, Communications, and Utilities	0.015	0.064	551				
Wholesale and Retail Trade	0.012	0.076	2351				
Services (Hotel, Business, Health)	0.018	0.074	897				
Services (Legal, Educational, Social)	0.026	0.099	700				
Public Administration and Others	0.069	0.118	7				

This table reports summary statistics for firm characteristics. The sample consists of 16,145 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021. Panel A presents firm-level summary statistics. Panel B presents *AI mentions* by one-digit SIC code, where *AI adoption* is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Descriptions of the other variables are provided in Appendix A.

4. AI adoption, investment efficiency, and firm value

4.1. Measure of investment efficiency

We measure investment efficiency following Biddle et al. (2009), a specification widely used in empirical studies of corporate investment decisions (e.g., Cheng et al., 2013; García Lara et al., 2016; Harford et al., 2017; Gao et al., 2021; Chay et al., 2023). In this framework, investment inefficiency is the deviation of a firm's actual investment from its predicted level, based on industry-year sales growth. For each industry-year group, defined using the Fama–French 48 industry classification and requiring at least 20 observations, we estimate $Investment = \beta_0 + \beta_1 Sales\ growth_{i,t-1} + \varepsilon_{it}$, where i denotes the firm and t the year. Following Biddle et al. (2009), we define investment as the sum of capital expenditures, R&D expenses, and acquisition expenditures, scaled by total assets.⁶ This definition is consistent with subsequent investment efficiency research, captures key components of firms' growth-related expenditures, and preserves the interpretation of residual deviations from predicted investment as over- or underinvestment. *Sales growth* is the firm's prior-year sales growth rate. The residuals ε_{it} capture deviations from the expected investment level implied by growth opportunities: positive residuals indicate higher-than-expected investment (potential overinvestment), whereas negative residuals indicate lower-than-expected investment (potential underinvestment). To obtain an overall measure of investment efficiency, we take the absolute value of the residuals and multiply by -1 for interpretability, so that higher values correspond to greater investment efficiency.

4.2. *AI mentions* and investment efficiency

We next examine whether AI adoption—as reflected in *AI mentions*—is associated with investment efficiency. Panel A of Table 3

⁶ As a robustness check, we also construct an alternative investment measure that subtracts the sale of property, plant, and equipment and scales by lagged total assets. The results in Online Appendix Table A.2 are broadly consistent with our main findings.

reports OLS regressions in which the dependent variable is *Investment efficiency*. The specifications include firm characteristics that prior research identifies as relevant for firms' investment policies (Richardson, 2006; Biddle et al., 2009). In column (1), we control for firm size, firm age, growth opportunities (book-to-market ratio), leverage, a dividend payer indicator, sales volatility, and investment volatility. Column (2) adds asset tangibility, the Altman (1968) Z-score, loss frequency, cash flow volatility, the ratio of cash flow to sales, the ratio of cash and short-term investments to PPE, and the operating cycle. Column (3) further incorporates governance variables: institutional ownership, a CEO duality indicator, board size, and the ratio of outside directors to board size. These controls capture multiple dimensions of firm risk, including business, financial, and governance risks, helping mitigate concerns that the results are driven by omitted risk factors. All regressions include firm and year fixed effects. In terms of magnitude, a one-standard-deviation increase in *AI mentions* (0.137) corresponds to a 2.17% increase in investment efficiency when evaluated at the mean of 0.101 and a 3.65% increase when evaluated at the median of 0.060.⁷

These OLS estimates motivate a closer examination of potential endogeneity. Unobservable firm characteristics could jointly affect investment efficiency and managers' emphasis on AI in earnings calls. To address this concern, we employ an instrumental variables (IV) approach. In the spirit of Chen and Srinivasan (2024), we construct an instrumental variable for *AI mentions* based on textual similarity between non-corporate academic AI publications and detailed 2017 NAICS industry descriptions.⁸ Specifically, we measure *AI technology exposure* as the normalized score derived from this similarity, using time-varying academic publications and fixed industry descriptions, to capture the extent to which each industry is exposed to developments in AI research. The relevance condition relies on the idea that the AI exposure score—derived from the volume and content of academic AI research linked to each industry—predicts firm-level AI adoption and managers' articulation of AI as a strategic priority. This association is estimated in the first-stage regression. Because the instrument is constructed exclusively from academic publications authored by university researchers—with all corporate coauthorships filtered out—it captures knowledge spillovers that are external to the corporate sector. This non-corporate origin is central to the exclusion argument: since the underlying research is produced by academics with no direct stake in firm-level investment decisions, the instrument reflects technological supply conditions in the broader research environment rather than demand-side factors specific to individual firms. Accordingly, the instrument is unlikely to influence unobserved firm-specific shocks to investment efficiency other than through firms' AI-related activities, supporting the exclusion restriction. While these design choices help mitigate common threats to IV identification, the exclusion restriction is fundamentally untestable and may not hold under certain conditions.⁹ We therefore interpret the 2SLS estimates as complementary to, rather than a substitute for, the OLS and mechanism analyses, and view them as part of a broader body of evidence rather than as standalone causal estimates.

Panel B of Table 3 presents the first- and second-stage results of the 2SLS estimation. In the first stage, AI technology exposure is positively and significantly associated with *AI mentions* at the 1% level, and the Cragg and Donald (1993) *F*-statistic of 118.83 exceeds the Stock–Yogo (2005) critical value at the 5% level, suggesting that weak instruments are unlikely to be a concern.¹⁰ In the second stage, reported in column (2), we regress *Investment efficiency* on the instrumented *AI mentions* and the full set of controls. The coefficient on instrumented *AI mentions* is positive and significant at the 5% level, indicating that the positive association between AI adoption and investment efficiency is unlikely to be fully attributable to endogeneity.

4.3. *AI mentions and firm value*

If AI adoption facilitates more efficient resource allocation—for example, by improving data use and internal decision-making processes—it may, over time, be reflected in higher firm value. We therefore examine whether firms with higher AI mentions experience subsequent changes in Tobin's *q*. The timing of our analysis follows research on technology adoption and organizational adjustment, which suggests that performance effects accumulate over several years as firms adapt workflows, invest in complementary assets, and train employees (e.g., Brynjolfsson and Hitt, 2000; Bartel et al., 2007; Babina et al., 2024a). Accordingly, any association between *AI mentions* and firm value would be expected to emerge gradually.

Panel A of Table 4 presents the OLS estimates. Columns (1)–(3) use Tobin's *q* in years $t + 1$, $t + 2$, and $t + 3$, respectively. The coefficients on *AI mentions* are positive and similar in magnitude across these horizons, suggesting a positive association between AI mentions and subsequent firm value over the three-year window. Columns (4) and (5), which use Tobin's *q* aggregated over two- and

⁷ The 2.17% is computed as $(0.137 \times 0.016)/0.101$.

⁸ While Chen and Srinivasan (2024) rely on patents to capture commercialized or strategically valuable AI activity, our use of academic AI research emphasizes conceptual and methodological advances and provides a broader view of AI's diffusion across industries, which is broadly consistent with our focus on the informational environment shaping executives' investment-related decisions.

⁹ The instrument's validity rests on the premise that academic AI research evolves in response to scientific opportunities rather than firm-specific investment decisions. Excluding corporate coauthorships removes the most direct channel through which firm-level activity could shape the instrument, but academic and corporate AI activity may share common drivers, such as broad shifts in the supply of AI-trained talent. In addition, because our instrument varies at the industry–year level by design (Chen and Srinivasan, 2024), identification exploits cross-industry variation in AI research intensity rather than within-industry firm-level shocks. We address these concerns in several ways: firm and year fixed effects absorb time-invariant firm characteristics and aggregate time trends; our control set captures observable firm-level heterogeneity; and we report weak-IV-robust inference using the tF procedure of Lee et al. (2022) and Anderson–Rubin confidence intervals. Online Appendix Table A.3 indicates that the main inferences are generally robust under these methods. Our conclusions do not rest solely on the 2SLS approach—the GDPR quasi-experiment in Section 6.2 provides an alternative identification strategy that exploits firm-level variation in regulatory exposure, and its findings are consistent with the IV results.

¹⁰ The smaller sample in Panel B reflects 390 firm-year observations with missing NAICS codes.

Table 3
AI Adoption and investment efficiency.

Panel A. OLS regression			
Independent variable	Investment efficiency		
	(1)	(2)	(3)
AI mentions	0.016* (1.777)	0.016* (1.748)	0.016* (1.731)
Firm size	0.004* (1.877)	0.005** (2.125)	0.004* (1.922)
Log(firm age)	0.013* (1.903)	0.011* (1.717)	0.010 (1.401)
Book-to-market	0.001 (0.477)	0.002 (0.583)	0.001 (0.452)
Leverage	0.026*** (2.970)	0.025** (2.571)	0.025** (2.574)
Dividend (indicator)	−0.006 (−1.512)	−0.006 (−1.467)	−0.006 (−1.469)
Sales volatility	0.008 (0.832)	0.008 (0.853)	0.008 (0.842)
Investment volatility	0.009 (0.701)	0.009 (0.446)	0.009 (0.444)
Asset tangibility		−0.003 (−0.315)	−0.003 (−0.334)
Z-score		−0.001 (−0.369)	−0.001 (−0.359)
Negative income (indicator)		−0.000 (−0.147)	−0.000 (−0.159)
Cash flow volatility		0.006 (0.260)	0.005 (0.241)
Cash flow to sales		−0.000 (−0.552)	−0.000 (−0.544)
Cash to PPE		−0.000*** (−2.783)	−0.000*** (−2.745)
Log(operating cycle)		0.001 (0.187)	0.001 (0.196)
Institutional ownership			−0.002 (−0.349)
CEO chair duality (indicator)			0.006** (1.967)
Board size			0.001 (1.314)
Board independence			0.024* (1.706)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Number of observations	15,901	15,901	15,901
Adj. R^2	0.362	0.363	0.363
Panel B. 2SLS regression			
	1st stage	2nd stage	
Independent variable	AI mentions	Investment efficiency	
AI technology exposure	(1)	(2)	
	0.005*** (5.649)		
AI mentions (instrumented)		0.300** (2.360)	
Control variables (as in column (3) of Panel A)			
Year fixed effects	Yes	Yes	
Firm fixed effects	Yes	Yes	
Number of observations	15,516	15,516	
Adj. R^2	0.227	–	
Cragg-Donald F -statistic	–	118.834	
Kleibergen-Paap Wald	–	31.915	

Panel A reports estimates from ordinary least squares (OLS) regressions in which the dependent variable is *Investment efficiency*, defined as the absolute value of the residuals from the investment model of Biddle et al. (2009), multiplied by -1 . Investment is the sum of capital expenditures, R&D expenses, and acquisition expenditures, scaled by total assets. The sample consists of 15,901 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021. AI adoption is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Panel B reports estimates from two-stage least squares (2SLS) regressions in which *AI technology exposure* serves as the instrumental variable for *AI mentions*. *AI technology exposure* is the normalized score based on textual similarity between university-authored AI publications and industry descriptions, capturing the extent to which each firm is exposed to developments in AI research. The dependent variables are *AI mentions* in column (1) and

Investment efficiency in column (2). Descriptions of the other variables are provided in Appendix A. T-values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

three-year horizons, also yield positive and significant coefficients, indicating that the association persists when Tobin's q is aggregated across multiple years.

Panel B of Table 4 presents the 2SLS estimates. The coefficients on instrumented *AI mentions* are positive and significant across all horizons, with larger magnitudes in years $t + 1$ and $t + 2$ and somewhat smaller, though still significant, estimates in year $t + 3$. The average effects over the two- and three-year horizons in columns (4) and (5) are similarly positive and significant, indicating an association between instrumented AI mentions and firm value that persists over multiple years. Overall, the 2SLS results are consistent with the view that a stronger managerial focus on AI, as reflected in earnings calls, is associated with higher firm value.¹¹

5. Potential mechanisms of AI adoption

5.1. Management sales forecast error

We focus on management sales forecast accuracy as a potential mechanism linking AI adoption to investment efficiency. AI may improve managerial forecasting by enabling the analysis of large-scale datasets and enhancing prediction beyond traditional human judgment. More accurate forecasts help firms better align investment with capacity and growth opportunities, reducing costly misallocations (Goodman et al., 2014; Tanaka et al., 2020).

In Panel A of Table 5, the dependent variable is *Management sales forecast error*, expressed as a percentage and measured as the absolute value of the difference between actual sales and the management sales forecast, divided by the management sales forecast.¹² The coefficients on *AI mentions* are consistently negative and significant across specifications that use different time windows (one-year sales, two-year average sales, and three-year average sales), indicating that higher AI adoption is associated with lower management sales forecast error and thus greater forecasting accuracy. Panel B of Table 5 presents the 2SLS estimates. The coefficient on instrumented *AI mentions* is negative but insignificant for one-year sales and becomes negative and significant when sales are averaged over two- and three-year horizons, suggesting that higher AI adoption is associated with improved management sales forecast accuracy, which may in turn contribute to more efficient investment decisions.¹³

To shed further light on this mechanism, we examine whether AI adoption is associated with changes in analyst sales forecast accuracy. Because analysts rely mainly on public information and management disclosures, whereas managers can use internal AI tools to refine proprietary forecasts, AI adoption is likely to affect managerial forecasts more directly than analyst forecasts. Consequently, any improvements in analyst forecast accuracy are likely to be indirect and limited. Reestimating the OLS regressions in Table 5, using *Analyst sales forecast error*—measured as the absolute value of the difference between actual sales and analyst forecasts, divided by analyst forecasts—as the dependent variable, we find that the coefficient on *AI mentions* is statistically insignificant across all time windows (Online Appendix Table A.5). Taken together, this evidence is consistent with the view that AI adoption primarily improves internal decision-making rather than the broader information set available to external parties.

5.2. Financial reporting quality

We next investigate whether AI adoption is associated with investment efficiency through financial reporting quality. AI-enabled systems may constrain earnings management by automating transaction monitoring and anomaly detection, raising the visibility of accounting irregularities and the expected cost of opportunistic manipulation. Prior studies show that higher-quality financial reporting is associated with fewer incidences of over- and underinvestment (Richardson, 2006; Biddle et al., 2009).

Table 6 presents the results. In columns (1) and (2), we estimate OLS regressions where the dependent variables are real earnings management and accrual-based earnings management, respectively, each multiplied by -1 so that higher values indicate higher financial reporting quality. Real earnings management is calculated as abnormal discretionary expenses (multiplied by -1) plus abnormal production costs, following Cohen and Zarowin (2010). Accrual-based earnings management is the absolute value of discretionary accruals estimated using the Kothari et al. (2005) model. Both transformed measures proxy for higher financial reporting quality but capture distinct channels—operational versus accounting-based manipulation. Following Kothari et al. (2005) and Cohen and Zarowin (2010), we include controls for $\log(1 + \text{analyst coverage})$, analyst forecast dispersion, Big Four auditor (indicator), and

¹¹ We also examine firm performance measured by stock returns. As reported in Online Appendix Table A.4, the coefficient on *AI mentions* is generally positive, but statistical significance is limited: annual stock returns are significantly positive in years $t + 1$ and $t + 2$, whereas the coefficient in year $t + 3$ is negative and not statistically significant. The average effects over multi-year horizons are positive and sometimes significant, but their magnitudes and significance levels are smaller and less consistent than those for Tobin's q in the 2SLS regressions. Taken together, these results indicate that although *AI mentions* is associated with a persistent increase in firm value as reflected in Tobin's q , equity markets may incorporate expected benefits relatively quickly, so that the association with realized stock returns is weaker and less stable over longer horizons.

¹² We match our data with the I/B/E/S Guidance database to obtain management sales forecasts and actual sales, requiring that forecast announcement dates precede the corresponding actual results announcements.

¹³ In additional tests, firm-level risk measures provide little evidence that AI adoption is associated with lower firm risk, suggesting that improved managerial forecast accuracy is unlikely to be driven by a shift toward safer projects (Online Appendix Table A.6).

Table 4
AI adoption and firm performance.

Panel A. OLS regression					
Independent variable	Tobin's q				
	Year $t + 1$ (1)	Year $t + 2$ (2)	Year $t + 3$ (3)	Average Year $t + 1$, Year $t + 2$ (4)	Average Year $t + 1$, Year $t + 2$, Year $t + 3$ (5)
AI mentions	0.946* (1.792)	0.942** (2.193)	0.949** (2.481)	0.900** (2.015)	0.813** (2.308)
Firm size	0.471*** (10.617)	0.136*** (2.962)	−0.059 (−1.330)	0.300*** (7.290)	0.188*** (4.988)
Asset tangibility	−0.459*** (−3.393)	−0.558*** (−3.885)	−0.361*** (−2.796)	−0.495*** (−3.851)	−0.441*** (−3.787)
ROA	−0.286 (−1.502)	0.257 (1.056)	0.064 (0.314)	−0.037 (−0.219)	−0.012 (−0.078)
Leverage	0.384** (2.099)	0.313 (1.587)	−0.037 (−0.159)	0.315* (1.882)	0.199 (1.254)
Sales growth	−0.001 (−1.417)	−0.001* (−1.738)	−0.000 (−1.020)	−0.001* (−1.794)	−0.001 (−1.564)
Sales volatility	0.093 (0.651)	−0.166 (−1.118)	−0.253 (−1.614)	−0.039 (−0.302)	−0.102 (−0.845)
Stock return volatility	4.439*** (2.864)	1.679 (1.075)	−4.015** (−2.321)	3.233** (2.472)	1.421 (1.279)
R&D intensity	1.756** (2.563)	2.337*** (3.201)	0.041 (0.063)	1.808*** (2.781)	1.297*** (2.643)
Institutional ownership	−0.196** (−2.148)	−0.149 (−1.556)	−0.217** (−2.068)	−0.171** (−2.006)	−0.173** (−2.048)
Board size	−0.036*** (−2.653)	−0.015 (−1.029)	−0.010 (−0.626)	−0.029** (−2.193)	−0.024** (−1.966)
Board independence	−0.403 (−1.592)	−0.259 (−1.044)	0.111 (0.418)	−0.325 (−1.470)	−0.189 (−0.924)
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	14,055	13,034	11,377	14,055	14,055
Adj. R^2	0.664	0.664	0.679	0.738	0.787
Panel B. 2SLS regression					
Independent variable	2nd stage dependent variable: Tobin's q				
	Year $t + 1$ (1)	Year $t + 2$ (2)	Year $t + 3$ (3)	Average Year $t + 1$, Year $t + 2$ (4)	Average Year $t + 1$, Year $t + 2$, Year $t + 3$ (5)
AI mentions (instrumented)	15.269*** (4.163)	14.566*** (4.025)	8.312*** (3.295)	14.688*** (4.261)	12.781*** (4.174)
Control variables (as in Panel A)	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Number of observations	13,736	12,735	11,121	13,736	13,736
Cragg-Donald F -statistic	103.881	99.571	138.102	103.881	103.881
Kleibergen-Paap Wald	26.890	24.448	27.691	26.890	26.890

Panel A reports estimates from ordinary least squares (OLS) regressions in which the dependent variable is Tobin's q , defined as the book value of assets plus market capitalization minus common equity minus deferred taxes and investment tax credit, scaled by total assets, measured over the period indicated in each column. The sample consists of 14,055 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021. Panel B reports second-stage estimates from two-stage least squares (2SLS) regressions in which *AI technology exposure* serves as the instrumental variable for AI adoption (*AI mentions*). *AI technology exposure* is the normalized score based on textual similarity between university-authored AI publications and industry descriptions, capturing the extent to which each firm is exposed to developments in AI research. *AI mentions* is defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Descriptions of the other variables are provided in Appendix A. T -values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

large security issuance (indicator), as well as common firm characteristics (firm size, leverage, book-to-market ratio, Z-score, institutional ownership, and board governance) that are relevant for financial reporting quality and that we use throughout the paper. In the OLS specifications, the coefficient on *AI mentions* is positive and significant at the 10% level for real earnings management and positive but insignificant for accrual-based earnings management, providing only limited evidence of an association between *AI mentions* and these measures of reporting quality. In columns (3) and (4), we report the 2SLS estimates after instrumenting for *AI*

Table 5

AI adoption and management sales forecast error as a channel to investment efficiency.

Panel A. OLS regression			
Independent variable	Management sales forecast error (%)		
	Sales Year $t + 1$	Average sales Year $t + 1$, Year $t + 2$	Average sales Year $t + 1$, Year $t + 2$, Year $t + 3$
	(1)	(2)	(3)
AI mentions	−2.819** (−2.024)	−3.397** (−2.518)	−3.340*** (−2.598)
Firm size	−0.079 (−0.285)	−0.026 (−0.095)	0.095 (0.355)
Log(firm age)	−0.583 (−0.534)	−1.289 (−1.126)	−1.456 (−1.224)
Book-to-market	0.107 (0.269)	0.128 (0.343)	0.141 (0.395)
Sales growth	−0.067*** (−2.841)	−0.068*** (−2.875)	−0.045*** (−2.624)
Leverage	0.990 (0.827)	0.654 (0.565)	0.688 (0.604)
ROA	−12.006*** (−3.995)	−10.270*** (−3.663)	−8.744*** (−3.364)
Stock return	−0.003 (−0.134)	−0.012 (−0.928)	−0.013 (−1.143)
Stock return volatility	−11.870 (−0.764)	−12.443 (−0.857)	−8.855 (−0.640)
Cash flow to sales	0.898** (2.023)	0.632** (2.087)	0.459* (1.733)
R&D intensity	−6.153 (−1.137)	−3.747 (−0.772)	−2.194 (−0.496)
Institutional ownership	−0.105 (−0.102)	−0.005 (−0.005)	0.097 (0.113)
Board size	−0.247* (−1.942)	−0.270** (−2.191)	−0.265** (−2.254)
Board independence	0.461 (0.241)	1.278 (0.685)	1.395 (0.769)
Top management ability index	0.188 (0.533)	0.189 (0.535)	0.119 (0.351)
Log(1+analyst coverage)	−0.513 (−1.602)	−0.519* (−1.753)	−0.430 (−1.486)
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Number of observations	5107	5260	5367
Adj. R^2	0.276	0.322	0.350
Panel B. 2SLS regression			
2nd stage dependent variable: Management sales forecast error (%)			
Independent variable	Sales Year $t + 1$	Average sales Year $t + 1$, Year $t + 2$	Average sales Year $t + 1$, Year $t + 2$, Year $t + 3$
	(1)	(2)	(3)
AI mentions (instrumented)	−41.554 (−1.487)	−43.209* (−1.694)	−43.186* (−1.650)
Control variables (as in Panel A)	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes
Number of observations	4947	5098	5204
Cragg-Donald F -statistic	17.097	20.797	19.863
Kleibergen-Paap Wald	6.375	7.794	7.417

Panel A reports estimates from ordinary least squares (OLS) regressions in which the dependent variable is *Management sales forecast error*, measured as the absolute value of the difference between actual sales and the management sales forecast, divided by the management sales forecast and expressed as a percentage; actual sales and sales forecasts are measured over the period indicated in each column. The sample consists of 5367 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021. Information on management sales forecasts and actual sales results is obtained from the I/B/E/S Guidance database, and the announcement date of the forecast value is required to be no later than the announcement date of the corresponding actual value. AI adoption is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Panel B reports second-stage estimates from two-stage least squares (2SLS) regressions in which *AI technology exposure* serves as the instrumental variable for *AI mentions*. *AI technology exposure* is the normalized score based on textual similarity between university-authored AI publications and industry descriptions, capturing the extent to which each firm is exposed to developments in AI research. Descriptions of the other variables are provided in Appendix A. T -values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6

AI adoption and financial reporting quality as a channel to investment efficiency.

Independent variable	OLS regression		2SLS regression	
	Real EM (1)	Accrual-based EM (2)	2nd stage dependent variable:	
			Real EM (3)	Accrual-based EM (4)
AI mentions	0.167* (1.646)	0.042 (0.925)		
AI mentions (instrumented)			5.203*** (3.582)	1.300** (2.195)
Firm size	0.046** (2.000)	0.038*** (4.129)	−0.014 (−0.453)	0.022* (1.839)
Book-to-market	0.105*** (2.726)	0.042** (2.205)	0.092** (2.325)	0.038** (1.967)
Asset tangibility	−0.186** (−2.138)	0.109* (1.894)	−0.184* (−1.889)	0.110* (1.846)
Leverage	0.006 (0.064)	0.047 (1.072)	−0.079 (−0.766)	0.024 (0.497)
Sales volatility	−0.173* (−1.662)	0.033 (0.682)	−0.090 (−0.814)	0.052 (0.978)
Cash flow volatility	−0.085 (−0.514)	0.081 (1.018)	−0.136 (−0.833)	0.066 (0.843)
Z-score	−0.060** (−2.220)	0.023** (2.453)	−0.040 (−1.480)	0.027*** (2.663)
Log(1+analyst coverage)	0.011*** (3.137)	−0.001 (−0.863)	0.012*** (2.898)	−0.001 (−0.694)
Analyst forecast dispersion	−0.570* (−1.722)	−0.235 (−1.013)	−0.692* (−1.826)	−0.278 (−1.147)
Big Four auditor (indicator)	0.047 (0.831)	0.078** (2.385)	0.065 (1.053)	0.083** (2.405)
Large security issuance (indicator)	0.056*** (2.836)	−0.008 (−0.805)	0.070*** (3.161)	−0.004 (−0.444)
Institutional ownership	0.076 (0.947)	0.017 (0.470)	0.093 (1.018)	0.021 (0.547)
CEO chair duality (indicator)	0.006 (0.215)	−0.011 (−0.842)	0.023 (0.729)	−0.004 (−0.314)
Board size	0.008 (1.025)	0.005 (1.173)	0.006 (0.685)	0.004 (0.956)
Board independence	−0.180 (−1.458)	−0.034 (−0.482)	−0.237* (−1.676)	−0.054 (−0.719)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Number of observations	12,456	12,422	12,144	12,110
Adj. R^2	0.272	0.254	-	-
Cragg-Donald F -statistic	-	-	80.169	80.094
Kleibergen-Paap Wald	-	-	23.679	23.689

Columns (1) and (2) report estimates from ordinary least squares (OLS) regressions. Columns (3) and (4) report second-stage estimates from two-stage least squares (2SLS) regressions in which *AI technology exposure* serves as the instrumental variable for AI adoption (*AI mentions*). *AI mentions* is defined as the ratio of AI-related keyword counts ([Appendix C](#)) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. *AI technology exposure* is the normalized score based on textual similarity between university-authored AI publications and industry descriptions, capturing the extent to which each firm is exposed to developments in AI research. The sample consists of 12,456 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021. In columns (1) and (3), the dependent variable is *Real EM*, calculated by multiplying abnormal discretionary expenses by -1 and adding abnormal production costs, following [Cohen and Zarowin \(2010\)](#). In columns (2) and (4), the dependent variable is *Accrual-based EM*, measured as the absolute value of discretionary accruals estimated using the [Kothari et al. \(2005\)](#) model. For interpretability, both measures are multiplied by -1 so that higher values indicate higher financial reporting quality. Descriptions of the other variables are provided in [Appendix A](#). T -values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

mentions. The coefficient on instrumented *AI mentions* is positive and statistically significant for both real and accrual-based earnings management, suggesting that firms with greater AI adoption tend to engage in less real or accrual-based earnings management.¹⁴

5.3. Patent activity

We also examine whether AI adoption is associated with firm-level patenting outcomes. AI may help managers better assess new projects and screen potential investments, lowering the cost of experimentation and enabling firms to commit capital to higher-quality opportunities (Cockburn et al., 2018). Process innovations, which involve new methods of production, can facilitate adjustments in firms' input mix and protect firm value, while product innovations expand firms' investment opportunity set (Bena et al., 2022; Babina et al., 2024a).

Table 7 presents the results. Columns (1) and (2) report OLS regressions in which the dependent variables are *Process patents to sales* and *Product patents to sales*, defined as the ratio of the number of process and product patents, respectively, to sales.¹⁵ Because these ratios scale patent counts by firm size, they capture innovation intensity rather than the level of innovative output. Process (product) patents are identified as those for which over 70% of claims are classified in the respective category, following Ganglmair et al. (2022). The regressions control for *R&D intensity* and other variables that are likely to affect firm innovation outcomes and capital allocation. The coefficients on *AI mentions* are positive and significant in both columns, indicating that higher AI adoption is associated with greater process and product patenting intensity. Columns (3) and (4) present the 2SLS estimates, which yield similar inferences regarding the association between *AI mentions* and patenting intensity.

Taken together, these results suggest that increased process and product patenting intensity is one channel through which AI adoption may be associated with more effective capital allocation and, potentially, improved firm performance.

6. Additional tests

6.1. Early versus later AI adoption periods

We examine whether the association between AI adoption and investment efficiency differs across earlier and later periods. Previous work suggests that firms may not immediately benefit from AI adoption, as they need time to establish new processes and invest in complementary intangible assets such as digital skills and technology readiness (Brynjolfsson et al., 2021). At the same time, broader AI adoption in later periods could attenuate any positive association with investment efficiency, as competition for AI talent and technology intensifies and stakeholders become more aware of AI's potential biases and limitations, which may foster skepticism about AI-based tools (D'Acunto et al., 2019). To explore these possibilities, we split the sample into an early period (2010–2015) and a later period (2016–2021) and reestimate the regressions in column (3) of Table 3 for each subsample. Table 8 reports the results. In the OLS regressions (columns (1) and (2)), the coefficient on *AI mentions* is positive and significant only in the early period. In the 2SLS specifications (columns (3) and (4)), the coefficients on instrumented *AI mentions* are not significant in either period. Overall, these results provide only weak evidence that any positive association between AI adoption and investment efficiency is concentrated among early adopters.

6.2. GDPR and AI-related investment efficiency

To assess how regulation may influence the association between AI adoption and investment efficiency, we focus on the GDPR in the EU, which has been in effect since May 25, 2018. The GDPR sets strict standards for handling personal data and applies to both EU-based and non-EU firms that target or monitor individuals in the EU, potentially constraining the use of data-driven AI in decision-making (Aridor et al., 2023).¹⁶

We implement a difference-in-differences design comparing treatment firms with significant EU exposure to control firms without such exposure. Treatment firms are those in the top 30% of average EU sales from 2015 to 2017 in the FactSet Revere Geographic Revenue database, and control firms are matched one-to-one using propensity scores from a logistic model that includes the variables in column (3) of Panel A of Table 3 and industry fixed effects, yielding 2755 firm-year observations for 296 treatment and 296 control firms over 2016–2021 (excluding 2018). We hypothesize that, after the GDPR, treatment firms experience a weaker association between AI adoption and investment efficiency if regulatory frictions constrain AI use.

¹⁴ In untabulated tests, standard operating efficiency metrics (operating expenses, sales per worker, and revenue-based productivity measures) provide little systematic evidence of operating efficiency gains. This finding is broadly consistent with recent work suggesting that early AI adoption is more closely associated with growth and product innovation than with operating efficiency (Babina et al., 2024a).

¹⁵ Unlike Babina et al. (2024a), who analyze levels and log changes of product and process patent counts, we study patenting intensity (patents to sales) rather than patent levels. As a robustness check, we reestimate columns (1) and (2) using Poisson models with the number of process and product patents as the dependent variables. For process patents, we obtain qualitatively similar results, whereas the coefficients on product patents are not statistically significant.

¹⁶ Firms with substantial EU exposure must adjust their data practices (for example, data minimization, explicit consent, and limits on certain forms of algorithmic decision-making), which can restrict data access, raise compliance costs, and narrow the scope of AI applications, potentially dampening AI-related efficiency gains.

Table 7

AI adoption and patenting activity as a channel to investment efficiency.

Independent variable	OLS regression		2SLS regression	
			2nd stage dependent variable:	
	Process patents to sales (1)	Product patents to sales (2)	Process patents to sales (3)	Product patents to sales (4)
AI mentions	0.275* (1.770)	0.338** (1.971)		
AI mentions (instrumented)			4.663* (1.660)	6.524* (1.789)
Firm size	0.060 (0.322)	0.037 (0.138)	0.031 (0.168)	−0.005 (−0.019)
Log(firm age)	0.096 (0.309)	0.342 (0.790)	0.124 (0.391)	0.381 (0.853)
Book-to-market	−0.448 (−0.383)	−0.602 (−0.370)	−0.386 (−0.332)	−0.514 (−0.320)
Asset tangibility	0.184 (0.386)	0.400 (0.598)	0.148 (0.311)	0.351 (0.529)
Leverage	−0.934* (−1.691)	−0.720 (−0.859)	−0.969* (−1.691)	−0.770 (−0.891)
R&D intensity	0.769 (0.634)	1.625 (0.921)	0.755 (0.623)	1.607 (0.913)
Stock return	−0.007 (−1.179)	−0.004 (−0.603)	−0.008 (−1.237)	−0.005 (−0.695)
Stock return volatility	0.687 (1.034)	0.644 (0.932)	0.687 (1.028)	0.642 (0.925)
Sales volatility	0.000* (1.713)	0.000 (1.593)	0.000* (1.694)	0.000 (1.562)
Cash to PPE	−11.942* (−1.683)	−12.163 (−1.564)	−11.536* (−1.654)	−11.592 (−1.528)
Institutional ownership	2.127 (1.364)	3.866* (1.707)	2.048 (1.348)	3.753* (1.693)
Board size	0.034* (1.680)	0.044 (1.603)	0.030 (1.503)	0.037 (1.385)
Board independence	−0.794 (−0.948)	−0.243 (−0.240)	−0.971 (−1.062)	−0.489 (−0.444)
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Number of observations	4411	4411	4363	4363
Adj. R ²	0.539	0.294	-	-
Cragg-Donald F-statistic	-	-	123.445	123.445
Kleibergen-Paap Wald	-	-	19.112	19.112

Columns (1) and (2) report estimates from ordinary least squares (OLS) regressions. Columns (3) and (4) report estimates from two-stage least squares (2SLS) regressions in which *AI technology exposure* serves as the instrumental variable for AI adoption (*AI mentions*). *AI mentions* is defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. *AI technology exposure* is the normalized score based on textual similarity between university-authored AI publications and industry descriptions, capturing the extent to which each firm is exposed to developments in AI research. In columns (1) and (3), the dependent variable is *Process patents to sales*, defined as the ratio of the number of process patents to sales. In columns (2) and (4), the dependent variable is *Product patents to sales*, defined as the ratio of the number of product patents to sales. Process (product) patents are identified as those for which more than 70% of claims are classified in the respective category, following Ganglmair et al. (2022). The sample consists of 4363 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021; the smaller sample reflects the merge with patent data provided by Noah Stoffman and Bernhard Ganglmair. Descriptions of the other variables are provided in Appendix A. T-values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 9 reports the results. In column (1), the key variable is the triple interaction among *AI mentions*; *Treatment*, an indicator equal to one for treatment firms, and zero for control firms; and *GDPR Post-adoption*, an indicator equal to one for the post-GDPR years 2019–2021, and zero otherwise, with *Treatment* and *GDPR Post-adoption* absorbed by firm and year fixed effects. The coefficient on this triple interaction is negative and significant, indicating that the positive association between *AI mentions* and investment efficiency is weaker for treated firms after GDPR implementation. This pattern remains when we include the control variables from column (3) of Panel A of Table 3 in column (2). To examine dynamics, we interact *AI mentions* and *Treatment* with year indicators for the six years around the GDPR, omitting 2015 as the benchmark year. In columns (3) and (4), the post-GDPR triple-interaction coefficients are generally negative and significant (except for 2021), and the decline in the association between *AI mentions* and investment efficiency is most pronounced in the first two years after adoption. Online Appendix Figure A.2 shows that the year-specific coefficients are close

Table 8

AI adoption and investment efficiency: early versus later AI adoption.

Independent variable	OLS regression		2SLS regression: 2nd stage dependent variable	
	Early AI adoption period (2010–2015) Investment efficiency	Later AI adoption period (2016–2021)	Early AI adoption period (2010–2015)	Later AI adoption period (2016–2021)
	(1)	(2)	(3)	(4)
AI mentions	0.078* (1.731)	0.015 (1.606)		
AI mentions (instrumented)			−4.881 (−1.324)	0.214 (0.854)
Control variables (as in column (3) of Panel A of Table 3)	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Number of observations	8129	7534	7901	7387
Adj. R ²	0.358	0.402	-	-
Cragg-Donald F-statistic	-	-	9.216	11.218
Kleibergen-Paap Wald	-	-	2.747	8.055

Columns (1) and (2) report estimates from ordinary least squares (OLS) regressions in which the dependent variable is *Investment efficiency*, defined as the absolute value of the residuals from the investment model of Biddle et al. (2009), multiplied by −1. Investment is the sum of capital expenditures, R&D expenses, and acquisition expenditures, scaled by total assets. The sample consists of 8129 firm-year observations from the StreetEvents–Compustat–CRSP database for 2010–2015 in columns (1) and (3) and 7534 firm-year observations for 2016–2021 in columns (2) and (4). AI adoption is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Columns (3) and (4) report second-stage estimates from two-stage least squares (2SLS) regressions in which *AI technology exposure* serves as the instrumental variable for *AI mentions*. *AI technology exposure* is the normalized score based on textual similarity between university-authored AI publications and industry descriptions, capturing the extent to which each firm is exposed to developments in AI research. Descriptions of the other variables are provided in Appendix A. *T*-values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

to zero in the pre-GDPR period and become more negative after the regulation takes effect, consistent with the parallel-trends assumption underlying the design. We also examine whether the GDPR is associated with a shift in comparative advantage toward large, data-rich firms and find that the attenuation effect appears to be smaller for large incumbents with richer data resources (Online Appendix Table A.7).¹⁷

6.3. Controlling for industry-level regulatory and other changes

A potential concern is that our estimates might be confounded by time-varying, industry-level shocks, such as broad regulatory changes. Our main analyses focus on U.S. firms and include firm and year fixed effects, which control for time-invariant firm characteristics and common macroeconomic shocks. As reported in Online Appendix Table A.8, we re estimate all baseline OLS specifications using industry–year fixed effects in place of year fixed effects, thereby absorbing time-varying shocks at the industry level. The results are broadly similar across specifications, although the estimates for financial reporting quality are somewhat weaker. While this approach cannot rule out all remaining sources of bias, the overall evidence is not consistent with the idea that omitted regulatory shocks fully account for our findings.

6.4. AI subfields and investment efficiency

To examine which AI themes in earnings-call discussions are more closely associated with investment efficiency, we decompose the AI-related terms in Appendix C into five subfields corresponding to layers of the AI stack: (i) core machine-learning methods, (ii) natural-language technologies, (iii) computer-vision and perception applications, (iv) data and analytics tools, and (v) computing and hardware infrastructure, following existing AI taxonomies (Goodfellow et al., 2016; Russell and Norvig, 2024). Each subfield variable is defined as the ratio of AI-related keywords in that category to total transcript words, and we reestimate the OLS specification in column (3) of Panel A of Table 3 using these five measures. Online Appendix Table A.9 shows that only the coefficient on the computing and hardware infrastructure subfield is positive and significant in the full sample, the early AI adoption period, and the later period. The coefficient on the data and analytics tools subfield is positive and significant only in the initial period. Equality tests across subfields are not rejected in the full sample or in the later period, providing little evidence that any single subfield has a stronger association with investment efficiency in either sample. In the early period, however, the equality tests are rejected, indicating that

¹⁷ We thank an anonymous reviewer for suggesting this insightful analysis.

Table 9

AI adoption, investment efficiency, and the general data protection regulation (GDPR).

Independent variable	Investment efficiency			
	(1)	(2)	(3)	(4)
AI mentions: a	0.026	0.034	0.030	0.039
Treatment (indicator) × GDPR Post-adoption (indicator)	(0.801) −0.006 (−0.709)	(1.144) −0.009 (−0.998)	(0.626)	(1.179)
a × GDPR Post-adoption (indicator)	0.096 (1.471)	0.064 (1.460)		
a × Treatment (indicator)	0.000 (0.008)	−0.008 (−0.198)	0.051 (0.841)	0.048 (0.896)
a × Treatment (indicator) × GDPR Post-adoption (indicator)	−0.142** (−1.995)	−0.105* (−1.871)		
a × Year ₂₀₁₆ (indicator)			0.043 (0.845)	0.045 (1.170)
a × Year ₂₀₁₇ (indicator)			−0.018 (−0.261)	−0.019 (−0.347)
a × Year ₂₀₁₉ (indicator)			0.040 (0.558)	0.010 (0.198)
a × Year ₂₀₂₀ (indicator)			0.189** (2.162)	0.156** (2.063)
a × Year ₂₀₂₁ (indicator)			0.020 (0.345)	0.012 (0.258)
Treatment (indicator) × Year ₂₀₁₆ (indicator)			−0.025 (−1.482)	−0.025 (−1.596)
Treatment (indicator) × Year ₂₀₁₇ (indicator)			−0.027 (−1.345)	−0.028 (−1.538)
Treatment (indicator) × Year ₂₀₁₉ (indicator)			−0.023 (−1.429)	−0.026 (−1.648)
Treatment (indicator) × Year ₂₀₂₀ (indicator)			−0.028* (−1.724)	−0.030* (−1.945)
Treatment (indicator) × Year ₂₀₂₁ (indicator)			−0.017 (−1.063)	−0.019 (−1.232)
a × Treatment (indicator) × Year ₂₀₁₆ (indicator)			−0.050 (−0.849)	−0.053 (−1.038)
a × Treatment (indicator) × Year ₂₀₁₇ (indicator)			−0.053 (−0.690)	−0.060 (−0.867)
a × Treatment (indicator) × Year ₂₀₁₉ (indicator)			−0.153* (−1.886)	−0.132* (−1.933)
a × Treatment (indicator) × Year ₂₀₂₀ (indicator)			−0.295*** (−2.802)	−0.258*** (−2.607)
a × Treatment (indicator) × Year ₂₀₂₁ (indicator)			−0.080 (−1.104)	−0.070 (−1.022)
Control variables (as in column (3) of Panel A of Table 3)	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Number of observations	2755	2755	2755	2755
Adj. R ²	0.488	0.501	0.495	0.506

The table reports estimates from ordinary least squares (OLS) regressions in which the dependent variable is *Investment efficiency*, defined as the absolute value of the residuals from the investment model of Biddle et al. (2009), multiplied by -1 . Investment is the sum of capital expenditures, R&D expenses, and acquisition expenditures, scaled by total assets. The sample consists of 2755 firm-year observations (296 treatment and 296 control firms) from the StreetEvents–Compustat–CRSP database for 2016–2021, excluding 2018, the year in which the GDPR came into effect. Treatment firms (top 30% in average EU sales, 2015–2017) and control firms are matched using a propensity score estimated by logistic regression with industry fixed effects and three-year averages of the matching variables from column (3) of Panel A in Table 3. AI adoption is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. *Treatment (indicator)* equals one for treatment firms, and zero for control firms. *GDPR Post-adoption (indicator)* equals one for 2019–2021, and zero otherwise. *Year (indicator)* equals one for each indicated year, and zero otherwise. Descriptions of the other variables are provided in Appendix A. *T*-values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

data and analytics tools and computing and hardware infrastructure are more strongly associated with investment efficiency than the other subfields.

6.5. AI mentions and investment efficiency by investment type

We decompose total investment into two components: capex, scaled by total assets, and non-capex, which includes R&D expenses and acquisition expenditures, also scaled by total assets. R&D investments are often high-risk and long-term, and their payoffs are

difficult to evaluate because the resulting knowledge assets are intangible and only imperfectly reflected in financial statements. This complexity likely increases the demand for AI's predictive and analytical capabilities when assessing project prospects and allocating capital to R&D. To examine how AI adoption relates to different investment types, we calculate the absolute value of the residuals from the investment model separately for capex and non-capex and reestimate the regressions in column (3) of Panel A of Table 3 using these two measures as dependent variables (Online Appendix Table A.10). The estimates indicate that *AI mentions* is more strongly associated with higher efficiency in non-capex (R&D and acquisition) investment than in capex, consistent with a more pronounced role for AI in settings where project outcomes are highly uncertain and information is harder to verify. These results are consistent with prior evidence that AI adoption is associated with greater product innovation and firm growth (Babina et al., 2024a) and suggest that, in our setting, AI adoption is also linked to higher efficiency in R&D-related investment.

6.6. *AI mentions, underinvestment, and overinvestment*

We decompose suboptimal investment into underinvestment and overinvestment to examine whether AI adoption is differentially associated with these two forms of investment inefficiency. *Underinvestment* is measured by negative residuals from the investment model, and *Overinvestment* is measured by positive residuals. We reestimate the regressions from column (3) of Panel A of Table 3, replacing the dependent variable with *Underinvestment* and *Overinvestment* (Online Appendix Table A.11). The results suggest that *AI mentions* is positively associated with investment efficiency for firms prone to underinvestment. The coefficient for overinvestment is negative but insignificant.

6.7. *AI expertise as an alternative measure of AI adoption*

We use the intensity of AI talent within the high-skilled workforce, measured as the ratio of AI specialists to total high-skilled workers, as an alternative proxy for AI adoption. This measure is constructed from USProfile and Compustat using AI-related job titles defined in Babina et al. (2024a), and high-skilled workers are identified based on specific vocational preparation (SVP) levels of four or higher in the USProfile database. This ratio reflects the extent to which AI expertise is integrated among skilled employees who interact with senior management and provide direct input into decision-making. We re estimate the main OLS regressions using this alternative measure and find that the results are generally consistent with those reported in the main analysis (Online Appendix Table A.12). The estimates indicate that a higher share of AI talent within the workforce is associated with greater investment efficiency, improved management forecast accuracy, better financial reporting quality, and higher levels of innovation, particularly in process development. Moreover, the interaction between AI talent intensity and the GDPR treatment indicator remains negative and significant after the regulation, suggesting that GDPR is associated with a weaker positive association between AI adoption and investment efficiency.

7. Summary and conclusion

This study examines how AI adoption relates to investment efficiency in a setting where AI may enhance managerial forecasting, financial reporting quality, and innovation activity, yet also entails sizable implementation costs and risks. We proxy AI adoption using *AI mentions* in earnings calls, defined as the proportion of AI-related terminology in senior management remarks and responses during these calls, and find that this measure is positively associated with 10-K discussions of AI applications in operational decision-making, product development, and business expansion, suggesting that it captures economically meaningful variation in AI adoption.

We find that higher AI adoption is positively associated with greater investment efficiency, measured as firms' deviations from expected investment conditional on their growth opportunities. This association generally remains in 2SLS specifications that instrument *AI mentions* with exposure to non-corporate academic AI research, constructed from textual similarity between university-authored AI publications and detailed industry descriptions. While short-horizon stock returns do not appear to fully reflect the potential benefits of AI adoption, firms with higher *AI mentions* tend to exhibit higher Tobin's q over longer horizons, consistent with a gradual accumulation of complementary investments and organizational adjustments.

Additional analyses suggest several channels through which AI adoption may be associated with investment efficiency. Firms with higher *AI mentions* exhibit more accurate management sales forecasts, which may help align investment with capacity and growth opportunities and reduce costly misallocations. AI adoption is also associated with higher financial reporting quality and greater process and product innovation intensity, as captured by earnings management measures and patenting activity, respectively. At the same time, regulatory frictions, including data-privacy regulation, appear to attenuate the association between AI adoption and investment efficiency, suggesting that the benefits of AI may depend on the broader regulatory environment.

Overall, the evidence suggests that AI adoption is an increasingly relevant factor in how firms allocate capital, as reflected in investment efficiency. Our study underscores AI as a technology that may shape how firms implement their investment decisions, though the magnitude of its effects may depend on the regulatory and organizational environment in which firms operate

CRediT authorship contribution statement

Sheng-Syan Chen: Supervision, Conceptualization. **Jungmin Kim:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Shu-Cing Peng:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jempfin.2026.101730](https://doi.org/10.1016/j.jempfin.2026.101730).

Appendix A. variable definitions

This appendix provides detailed descriptions of the variables used in the tables.

Variable	Definition
AI mentions ($\times 1000$)	Ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. Source: StreetEvents database
Analyst forecast dispersion	Standard deviation of analyst earnings forecasts. Source: I/B/E/S
Asset tangibility	Ratio of net property, plant, and equipment (PPE) to total assets. Source: Compustat
Big Four auditor (indicator)	Indicator equal to one if the firm engaged a Big Four auditor, and zero otherwise. Source: Compustat
Board independence	Ratio of outside directors to the total number of directors on the board. Source: BoardEx
Board size	Number of directors. Source: BoardEx
Book-to-market	Ratio of the book value of equity to the market value of equity. Source: Compustat
Cash flow to sales	Ratio of cash flow from operations to total sales. Cash flow from operations includes income before extraordinary items, extraordinary items, discontinued operations, depreciation and amortization, deferred taxes, equity in a net loss of unconsolidated subsidiaries, losses from the sale of PPE, and other funds from operations. Source: Compustat
Cash flow volatility	Standard deviation of the ratio of cash flow from operations to total sales over the past five years, where cash flow from operations is defined as above. Source: Compustat
Cash to PPE	Ratio of cash and short-term investments to net PPE. Source: Compustat
CEO chair duality (indicator)	Indicator equal to one if the CEO is also the chair of the board, and zero otherwise. Source: BoardEx and ExecuComp
Dividend (indicator)	Indicator equal to one if a firm paid a dividend, and zero otherwise. Source: Compustat
Firm size	Natural logarithm of market capitalization. Source: Compustat
Institutional ownership	Ratio of the number of shares held by all institutional investors to the total number of common shares outstanding. Source: Thomson/Refinitiv 13F
Investment volatility	Standard deviation of investment over the past five years, where investment is defined as the ratio of the sum of capital expenditures, R&D expenses, and acquisition expenditures to total assets. Source: Compustat
Large security issuance (indicator)	Indicator equal to one if the firm's issued securities exceed 10% of total assets, and zero otherwise. Source: CRSP
Leverage	Ratio of the book value of debt to total assets. The book value of debt is the sum of long-term debt and debt in current liabilities. Source: Compustat
Log(firm age)	Natural logarithm of the number of years since the firm first appeared in Compustat. Source: Compustat
Log(operating cycle)	Natural logarithm of the operating cycle, where the operating cycle is the sum of the ratio of receivables to sales and the ratio of inventory to cost of goods sold, multiplied by 360. Source: Compustat
Negative income	Indicator equal to one if net income before extraordinary items is negative, and zero otherwise. Source: Compustat
R&D intensity	Ratio of R&D expenses to total assets. Source: Compustat
ROA	Ratio of operating income after depreciation to total assets. Source: Compustat
Sales growth	Annual growth of total sales. Source: Compustat
Sales volatility	Standard deviation of total sales over the past five years. Source: Compustat
Stock return	Buy-and-hold daily returns over the fiscal year. Source: CRSP
Stock return volatility	Standard deviation of daily returns over the fiscal year. Source: CRSP
Z-score	$3.3 \times (\text{pretax income} / \text{total assets}) + 1.0 \times (\text{total sales} / \text{total assets}) + 0.25 \times (\text{retained earnings} / \text{total assets}) + 0.5 \times (\text{working capital} / \text{total assets})$ (Altman, 1968). Source: Compustat

Appendix B. measuring firms' AI applications in earnings calls and 10-K business descriptions

B.1 Prompt for ChatGPT

You are an expert in artificial intelligence and machine learning, and your task is to assess the company's applications of AI-related technologies based on an excerpt from conference transcripts/10-K business descriptions. This excerpt features discussions among the company's executives regarding the utilization of various AI-related technologies. Please answer the following questions based solely on the information provided in the excerpt.

Question 1: In this excerpt, is the firm discussing its involvement in AI-related technologies?

- (a) Yes, it discusses the company's application or utilization of AI-related technologies.
- (b) No, it primarily discusses the company's investment in AI-related businesses.
- (c) No, it primarily discusses the company's strategies for AI-related operations.
- (d) No, it primarily discusses the importance of AI-related technologies.
- (e) No, it primarily discusses the company's acquisitions and potential business expansion in AI-related businesses.

Question 2–1: If your answer to Question 1 is (a) yes, specify the types of AI-related technology applications employed by the company using a maximum of five words.

Question 2–2: If your answer to Question 1 is (a) yes, specify the purposes for which the AI-related technology is employed using a maximum of five words.

Question 2–3: If your answer to Question 1 is (c) or (e), specify the areas of AI-related strategic focus using a maximum of five

words.

Artificial intelligence (AI) and machine learning encompass a wide array of technologies, frameworks, and applications. We choose keywords based on their significance in either the development or enhancement of AI or machine learning algorithms. The inclusion of specific keywords depends on their contextual relevance and applicability to the specific topic under discussion. Certain keywords, like ‘parallel processing,’ although instrumental in the realm of AI, may be excluded from our dictionary if their contextual relevance to a particular discussion is limited.

B.2 Identifying AI technology keywords in ChatGPT’s response to a comprehensive list of keywords

1. We excluded terms such as “artificial intelligence,” “AI,” “Machine learning,” and “ML.” However, keywords that contain artificial intelligence, AI, “Machine learning,” or ML are included.
2. A graphics processing unit (GPU) forms the core of AI. We exclude “GPU” as a standalone term but include keywords that contain “GPU.”
3. If a technology (e.g., stereo processing) is not inherently related to AI or ML but has the potential to incorporate AI and ML techniques, we exclude the specific keywords.

Appendix C. AI-related keywords used in AI dictionary

Keywords used	
3d computer vision	deep learning systems
accelerated computing	deep learning techniques
adas (advanced driver assistance systems)	deep learning technologies
advanced analytics	deep learning technology
ai and machine learning	deep learning toolkit
ai assistant	deep learning tools
ai automation	deep learning training
ai machine learning	deep learning vision technology
ai platform	deep neural net
ai platforms	deep neural network
ai processing	deep neural networks
ai technology	Deephealth
ai tools	drive px 2
ai/machine learning	edge computing
ai-driven software solutions	embedded computer vision
animal detection	end-to-end ai computing platform
artificial intelligence algorithms	enterprise ai platform
artificial intelligence analysis	ethos therapy system
artificial intelligence and machine learning	face detection
artificial intelligence capabilities	facial recognition
artificial intelligence engine	gpu architecture
artificial intelligence models	gpu computing
artificial intelligence platform	gpu deep learning
artificial intelligence pricing tool	gpu-accelerated
artificial intelligence processing	gpu-based artificial intelligence and machine learning
artificial intelligence services	Gpus
artificial intelligence software tools	high-performance computing
artificial intelligence solutions	image processing
artificial intelligence technologies	image recognition
artificial intelligence technology	industrial machine learning services
artificial intelligence tools	intelligent automation
audio ai	intelligent virtual assistant
augmented intelligence	machine intelligence
big data analytics	machine learning algorithm
big data applications	machine learning algorithms
caffe (a deep learning framework)	machine learning applications
chatbots	machine learning capabilities
classical machine learning	machine learning engine
cloud computing	machine learning frameworks
cloud service providers	machine learning inference
cognitive automation	machine learning methods
cognitive computing	machine learning models
cognitive learning	machine learning platform
cognitive tool kit	machine learning services
computer vision algorithms	machine learning software
computer vision analytics	machine learning techniques
computer vision capabilities	machine learning technologies
Appendix C – continued from the previous page	
Keywords used	

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Keywords used	
computer vision chip	machine learning technology
computer vision processing	machine learning tools
computer vision processors	machine vision
computer vision service	machine-vision
computer vision socs	natural language processing
computer vision solutions	natural language understanding
computer vision technology	neural nets
conversational ai	neural network
convolutional neural networks	neural networks
core ml	nvidia gpus
cuda (compute unified device architecture)	object detection
cv2 (computer vision)	object identification
cv22 (computer vision)	object recognition
cvflow	ontap ai
data science	Openvino
data scientists	predictive analytics
deep analytics	predictive modeling
deep brew	Pytorch
deep convolutional neural networks	self-driving car platform
deep learning ai applications	self-driving cars
deep learning algorithm	sensai software stack
deep learning algorithms	sensiml analytics toolkit
deep learning applications	smart analytics
deep learning capabilities	Spark
deep learning framework	speech synthesis
deep learning inference	Supercomputer
deep learning models	Supercomputers
deep learning networks	Supercomputing
deep learning platform	tensor core
deep learning recommender models	Tensorflow
deep learning software	tensorflow lite
deep learning stack	Tensorrt
deep learning supercomputer	voice user interface

This table lists AI-related keywords that occur at least ten times in earnings call transcripts. A comprehensive list of all keywords used in the analysis is available upon request.

Appendix D. excerpts from earnings call transcripts discussing AI applications

The “intensive margin” of AI benefits refers to the extent to which AI technologies may help firms select better projects from a given pool of potential investments. This encompasses improving the quality of project selection, enhancing the efficiency of the selection process, and optimizing resource allocation across existing projects.

“Using advanced cloud computing and machine learning algorithms, we are remotely identifying root causes and are being more prescriptive with our response.” (Excerpt from Diebold Nixdorf Inc.’s conference call on July 29, 2021)

“...innovating with advanced analytics to optimize key processes, for example, using machine learning for product allocation, pricing, or promotion planning.” (Excerpt from Tapestry Inc.’s conference call on October 30, 2018)

“Through our continued work in optimizing our machine learning delivery estimate models, strategically positioning inventory and leveraging partner relationships, as you can see from the top chart, we were able to increase the number of 2-day ship items by 198% in the first quarter year-over-year.” (Excerpt from Overstock.com Inc.’s conference call on April 30, 2020)

“This will replace the collection of disparate systems we use for inventory planning today and will infuse artificial intelligence and machine learning into both in-season product management and long lead time inventory planning.” (Excerpt from Duluth Holdings Inc.’s conference call on December 2, 2021).

The “extensive margin” of AI benefits refers to how AI expands the scope of a firm’s activities by discovering new opportunities, or increasing the pool of projects and strategies available to the firm.

“Today, through the Ulta Beauty app and ulta.com, we provide members with unique recommendations across a variety of experiences, including new products, products based on category preferences, reminders for replenishment or handpicked items, all powered by our internal AI platform.” (Excerpt from Ulta Beauty Inc.’s conference call on August 27, 2020)

“In addition, the material scientists on our innovation team are developing new ways to adapt Fortrex technology and expand its application across product lines by using artificial intelligence modeling.” (Excerpt from Cooper-Standard Holdings Inc.’s conference call on May 3, 2017)

“..., myWW Plus will deliver personalized content along their journey, powered by a smart personalization platform that uses machine learning and AI to identify the preferences of each unique member.” (Excerpt from WW International Inc.’s conference call on October 29, 2020)

“Finally, as we target doubling our direct connection to consumers, we’re ramping up investment in digital capabilities ranging from data science and analytics to machine learning to augmented reality to image recognition and personalization.” (Excerpt from

Nike Inc.'s conference call on September 26, 2017)

Appendix E. correlation matrix of AI adoption measure and 10-K business description disclosures

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) AI mentions	1.000						
(2) No. of intensive margin	0.234***	1.000					
(3) No. of extensive margin	0.241***	0.828***	1.000				
(4) Ratio of intensive margin	0.239***	0.878***	0.771***	1.000			
(5) Ratio of extensive margin	0.243***	0.672***	0.858***	0.781***	1.000		
(6) Ratio of litigious words	0.028***	0.066***	0.060***	0.061***	0.046***	1.000	
(7) Ratio of precise words	0.065***	0.072***	0.112***	0.070***	0.113***	0.032***	1.000

This table presents the correlation coefficients between our measure of a firm's AI adoption and its AI application disclosure language in 10-K business descriptions. AI adoption is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. *No. of intensive margin* is the total number of sentences in 10-K business descriptions that address how AI technologies help firms select better projects from the existing project pool. *No. of extensive margin* is the total number of sentences in 10-K business descriptions that relate to expanding the scope of activities, discovering new opportunities, or increasing the pool of options available to a firm. *Ratio of intensive margin* is the number of AI sentences that relate to the intensive margin of AI benefits to the total number of AI sentences containing keywords (Appendix C) in 10-K business descriptions (i.e., *AI sentences*). *Ratio of extensive margin* is the ratio of the number of sentences that relate to the extensive margin of AI benefits to the total number of *AI sentences*. *Ratio of litigious words* is the ratio of the number of litigious words to the total number of words in *AI sentences*. *Ratio of precise words* is the ratio of the number of precise words to the total number of words in *AI sentences*. Litigious and precise words are from the predefined word lists proposed by Loughran and McDonald (2011). *** denotes significance at the 1% level.

Appendix F. validity test of AI adoption measure: intensive versus extensive margin of AI benefits in 10-K disclosures

Independent variable	Intensive AI benefits (indicator) (1)	Extensive AI benefits (indicator) (2)
AI mentions	0.151*** (2.850)	0.130*** (2.683)
Firm size	0.010* (1.707)	0.015** (2.301)
Log(firm age)	0.047** (1.998)	−0.013 (−0.551)
Book-to-market	0.005 (0.994)	0.001 (0.159)
Leverage	0.008 (0.430)	0.029 (1.377)
Cash holdings	−0.022 (−0.664)	0.017 (0.592)
ROA	−0.018 (−1.426)	−0.017 (−1.360)
Stock return	0.000*** (2.809)	0.000*** (2.732)
Sales volatility	−0.035 (−1.550)	−0.025 (−1.124)
R&D intensity	−0.017 (−0.408)	0.041 (1.126)
Institutional ownership	0.006 (0.368)	0.000 (0.001)
Board size	0.004 (1.615)	0.005** (2.000)
Board independence	0.010 (0.293)	−0.074* (−1.952)
Log(1 + 10-K size)	0.004 (0.499)	0.002 (0.281)
Year fixed effects	Yes	Yes
Firm fixed effects	Yes	Yes
Number of observations	14,653	14,653
Adj. R ²	0.575	0.559

The table presents estimates from ordinary least squares (OLS) regressions. In column (1), the dependent variable is *Intensive AI benefits (indicator)*, which equals one if 10-K filings mention AI-related keywords or phrases describing how AI technologies assist firms in enhancing decision-making for existing business operations, and zero otherwise; examples include improving pricing strategies,

enhancing inventory management, and streamlining product evaluation. In column (2), the dependent variable is *Extensive AI benefits (indicator)*, which equals one if 10-K filings mention AI-related keywords or phrases indicating how AI technologies help firms expand their investment opportunities, and zero otherwise; examples include launching new products, expanding market reach, and identifying strategic partnerships. In Panel A, the sample consists of 14,881 firm-year observations from the StreetEvents–Compustat–CRSP database for the period 2010–2021. AI adoption is proxied by *AI mentions*, defined as the ratio of AI-related keyword counts (Appendix C) in senior management remarks and responses during earnings calls to the total number of words in the corresponding transcripts. *Cash holdings* is the ratio of cash and short-term investments to total assets. $\text{Log}(1 + 10\text{-K size})$ is the natural logarithm of one plus the total number of words in a firm's 10-K report. Descriptions of the other variables are provided in Appendix A. *T*-values in parentheses are based on standard errors adjusted for heteroscedasticity and clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

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